

Chapter 1

Motivation

Our main motivation throughout the book will be to understand the applications of ergodic theory to certain problems outside of ergodic theory, in particular to problems in number theory. As we will see, this requires a good understanding of particular examples, which will often be of an algebraic nature. Therefore, we will start with a few concrete examples, and state a few theorems arising from ergodic theory, some of which we will prove within this volume. In Sect. 1.8 we will discuss ergodic theory as a subject in more general terms⁽¹⁾.

1.1 Examples of Ergodic Behavior

The *orbit* of a point $x \in X$ under a transformation $T : X \rightarrow X$ is the set $\{T^n(x) \mid n \in \mathbb{N}\}$. The structure of the orbit can say a great deal about the original point x . In particular, the behavior of the orbit will sometimes detect special properties of the point. A particularly simple instance of this appears in the next example.

Example 1.1. Write $\mathbb{T}$ for the quotient group $\mathbb{R}/\mathbb{Z} = \{x + \mathbb{Z} \mid x \in \mathbb{R}\}$, which can be identified with a circle (as a topological space, this can also be obtained as a quotient space of $[0, 1]$ by identifying 0 with 1); there is a natural bijection between $\mathbb{T}$ and the half-open interval $[0, 1)$ obtained by sending the coset $x + \mathbb{Z}$ to the fractional part of x . Let $T : \mathbb{T} \rightarrow \mathbb{T}$ be defined by $T(x) = 10x \pmod{1}$. Then $x \in \mathbb{T}$ is rational if and only if the orbit of x under T is finite. To see this, assume first that $x = \frac{p}{q}$ is rational. In this case the orbit of x is some subset of $\{0, \frac{1}{q}, \dots, \frac{q-1}{q}\}$. Conversely, if the orbit is finite then there must be integers m, n with $1 \leq n < m$ for which $T^m(x) = T^n(x)$. It follows that $10^m x = 10^n x + k$ for some $k \in \mathbb{N}$, so x is rational.

Detecting the behavior of the orbit of a given point is usually not so straightforward. Ergodic theory generally has more to say about the orbit of

“typical” points, as illustrated in the next example. Write χ_A for the indicator function of a set,

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A. \end{cases}$$

Example 1.2. This example recovers a result due to Borel [40]. We shall see later that the map $T : \mathbb{T} \rightarrow \mathbb{T}$ defined by $T(x) = 10x \pmod{1}$ preserves Lebesgue measure m on $[0, 1)$ (see Definition 2.1), and is *ergodic* with respect to m (see Definition 2.13). A consequence of the pointwise ergodic theorem (Theorem 2.30) is that for any interval

$$A(j, k) = [\frac{j}{10^k}, \frac{j+1}{10^k}),$$

we have

$$\frac{1}{N} \sum_{i=0}^{N-1} \chi_{A(j,k)}(T^i x) \longrightarrow \int_0^1 \chi_{A(j,k)}(x) dm(x) = \frac{1}{10^k} \quad (1.1)$$

as $N \rightarrow \infty$, for almost every x (that is, for all x in the complement of a set of zero measure, which will be denoted a.e.). For any block $j_1 \dots j_k$ of k decimal digits, the convergence in (1.1) with $j = 10^{k-1}j_1 + 10^{k-2}j_2 + \dots + j_k$ shows that the block $j_1 \dots j_k$ appears with asymptotic frequency $\frac{1}{10^k}$ in the decimal expansion of almost every real number in $[0, 1]$.

Even though the ergodic theorem only concerns the orbital behavior of typical points, there are situations where one is able to describe the orbits for *all* starting points.

Example 1.3. We show later that the circle rotation $R_\alpha : \mathbb{T} \rightarrow \mathbb{T}$ defined by $R_\alpha(t) = t + \alpha \pmod{1}$ is *uniquely ergodic* if α is irrational (see Definition 4.9 and Example 4.11). A consequence of this is that for any interval $[a, b) \subseteq [0, 1) = \mathbb{T}$,

$$\frac{1}{N} \sum_{n=0}^{N-1} \chi_{[a,b)}(R_\alpha^n(t)) \longrightarrow b - a \quad (1.2)$$

as $N \rightarrow \infty$ for *every* $t \in \mathbb{T}$ (see Theorem 4.10 and Lemma 4.17). As pointed out by Arnol'd and Avez [7] this equidistribution result may be used to find the density of appearance of the digits⁽²⁾ in the sequence $1, 2, 4, 8, 1, 3, 6, 1, \dots$ of first digits of the powers of 2:

$$1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, \dots$$

A set $A \subseteq \mathbb{N}$ is said to have *density* $\mathbf{d}(A)$ if

$$\mathbf{d}(A) = \lim_{k \rightarrow \infty} \frac{1}{k} |A \cap [1, k]|$$